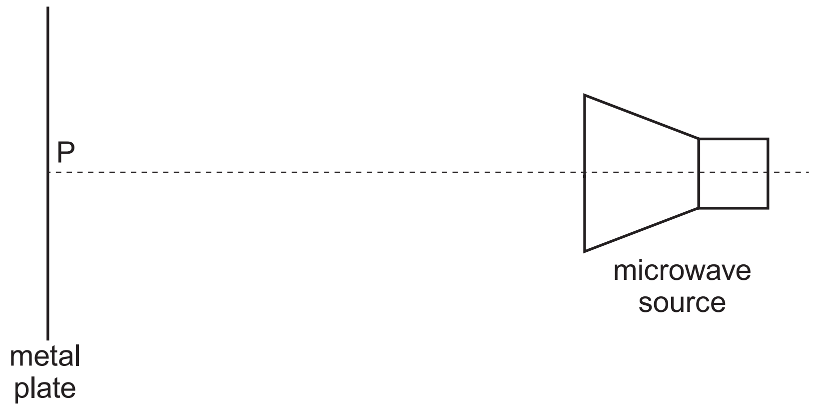
Examiner only

6. *(a)* In the set-up shown a series of *antinodes* is detected, at the following distances from the metal plate: 8 mm, 24 mm, 40 mm, 56 mm, 72 mm.



NOT TO SCALE

(i) Referring to the diagram, explain in terms of *interference* how an antinode is produced. [2]

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(ii) From the data, determine whether there is a node or an antinode at point **P** (on the metal plate). Give your reasoning in terms of wavelength. [2]

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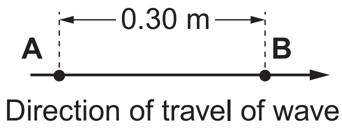
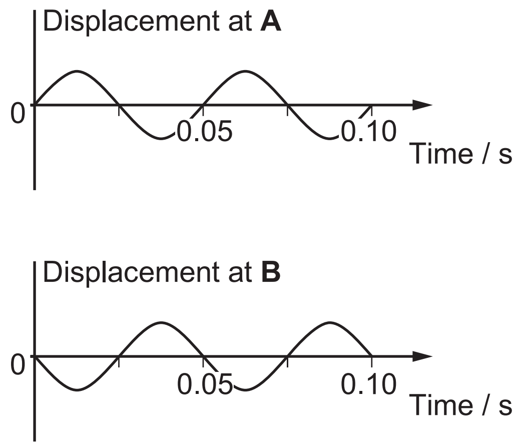
Examiner only

(b) Displacement–time graphs are given for two points, **A** and **B**, 0.30 m apart, in the path of a **progressive** wave.





(i) State the phase relationship between the displacements at **A** and **B**, then determine the longest, and the second longest, wavelength that the wave could have. [3]

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(ii) The speed of the waves is known to be between 10 m s^{-1} and 15 m s^{-1}. Determine which of the two wavelengths in *(b)*(i) is the correct one, giving your reasoning. [3]

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